Muons cosmiques

TP Césire

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LPSC, Grenoble

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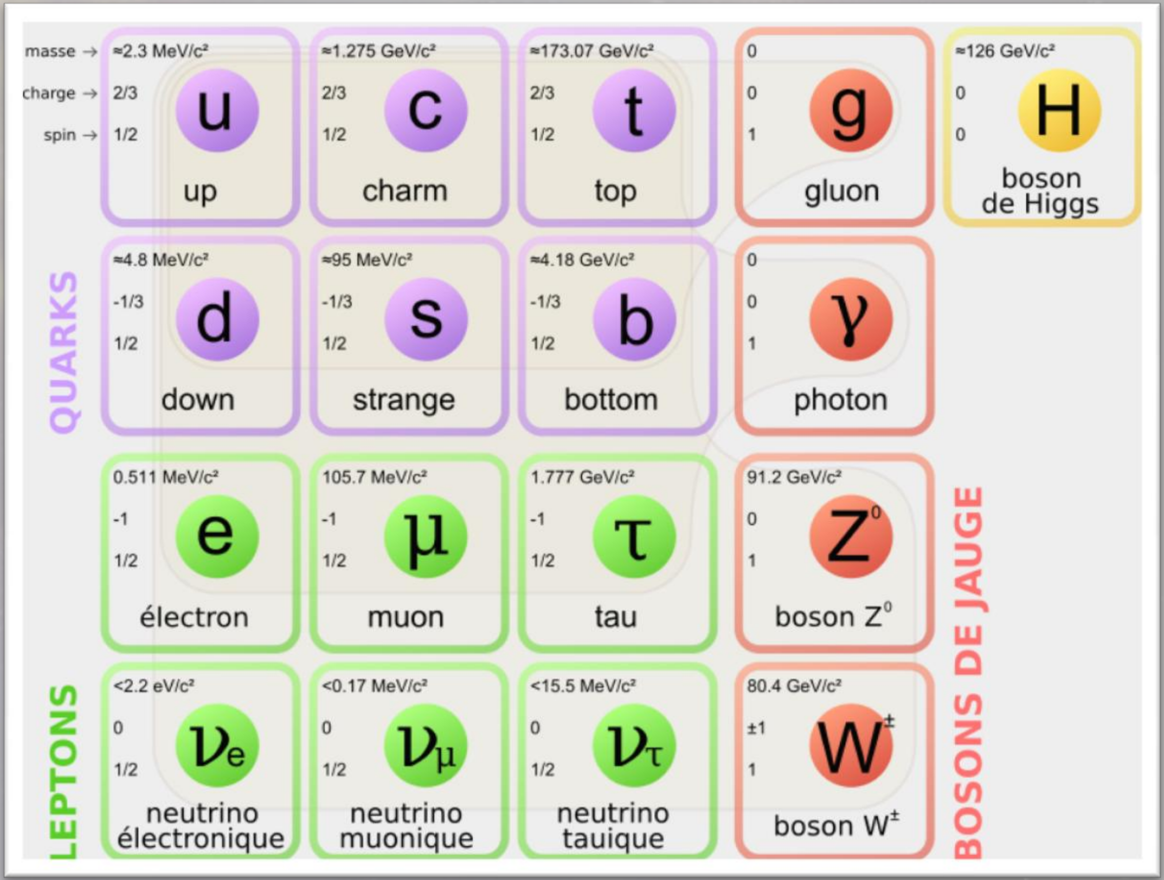
Le Modèle Standard de la physique des particules est une théorie qui concerne

- l'électromagnétisme
- les interactions nucléaires faible et forte
- la classification de toutes les particules subatomiques connues

Les quarks et les leptons sont des fermions fondamentaux – les particules qui constituent la matière.



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Classification	Fermion
Composition	*Élémentaire*
Groupe	Lepton
Génération	2ᵉ
Symbole	μ^-
Antiparticule	Antimuon
Propriétés physiques	
Masse	105,66 $MeV.c^{-2}$ ($1{,}88 \times 10^{-28}$ kg)
Charge électrique	$-1{,}60217653(14) \times 10^{-19}$ C
Charge de couleur	0
Spin	½
Durée de vie	2,2 μs / ~10^(-6) s
Historique	
Découverte	Carl D. Anderson, 1936

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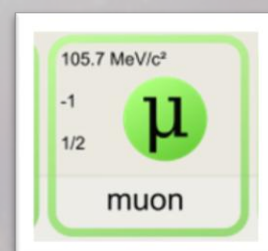
Les muons ont été découverts par Carl David Anderson à l'Institut de technologie de Californie, en 1936, alors qu'ils travaillaient sur les rayons cosmiques.



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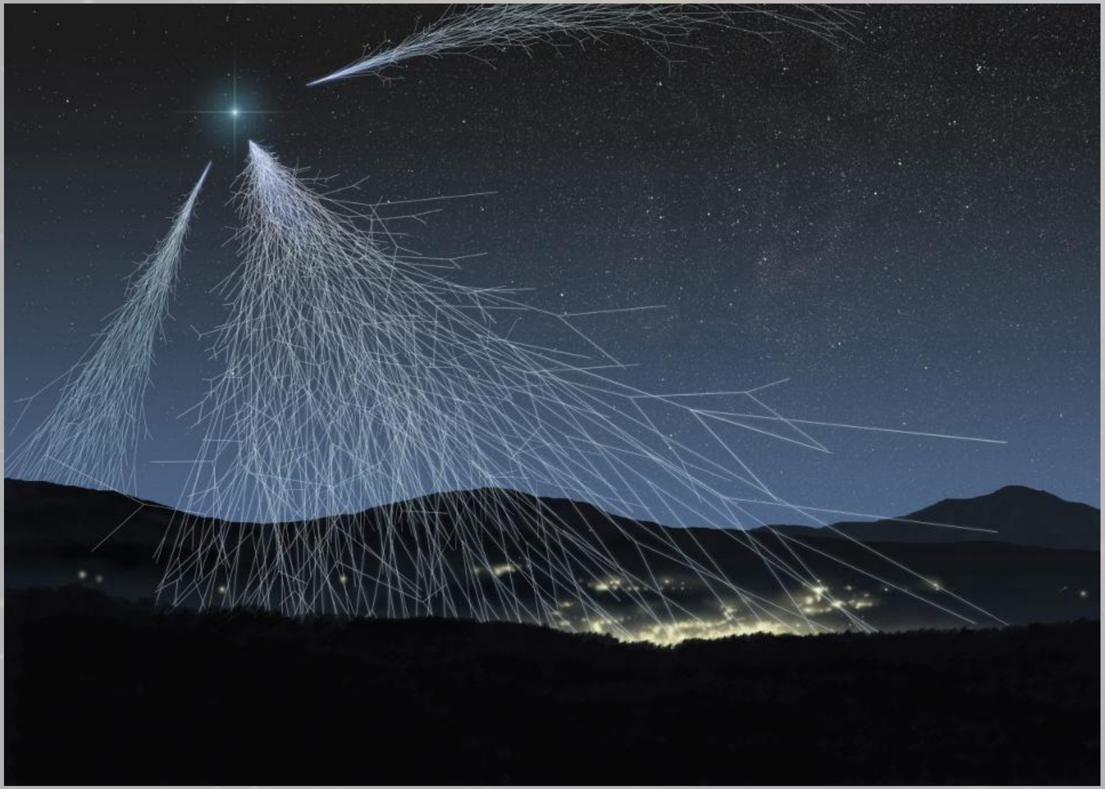
Le rayonnement cosmique est le flux de noyaux atomiques et de particules de haute énergie (c'est-à-dire relativistes) qui se propagent dans le vide interstellaire.

L’origine:
les phénomènes astronomiques très énergétiques comme l’explosion de supernovas, les galaxies actives, les trous noirs;
+ le vent solaire



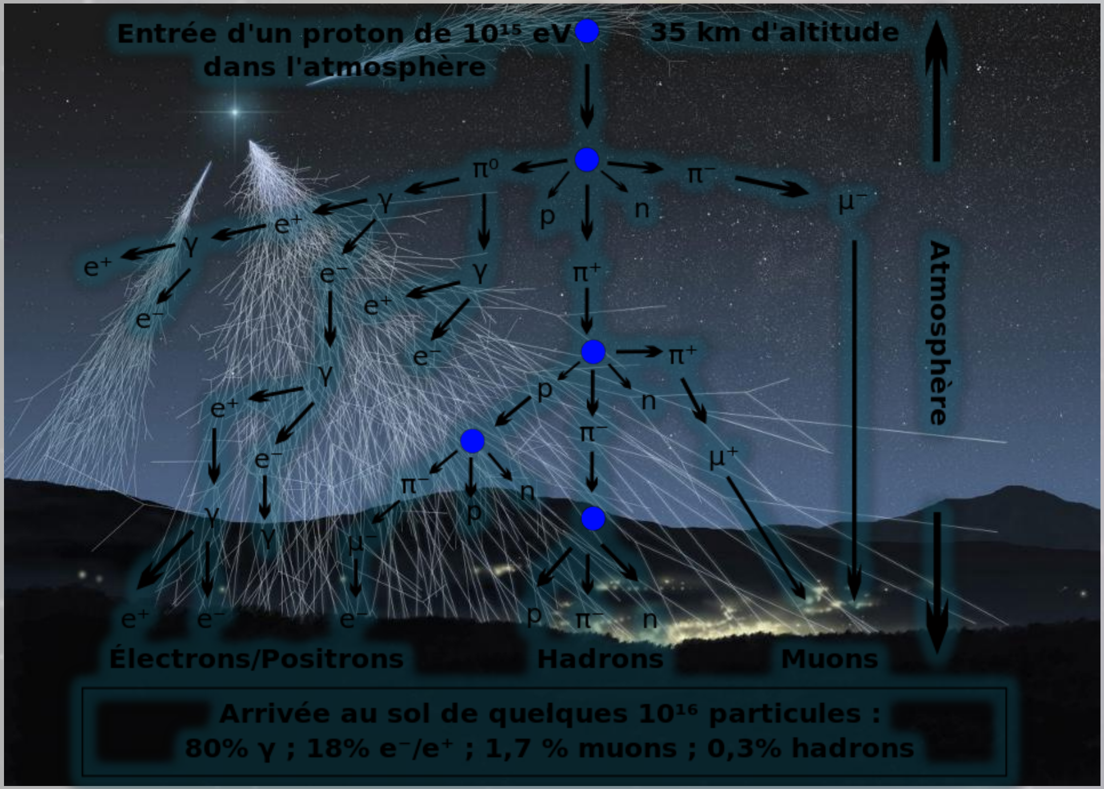
Le spectre:

principalement
- **des protons**
- **des rayons gammas**
- **des neutrinos**

et aussi
- **des noyaux d’hélium**
- **d’électrons**
- **des nucléons**

Dans l'atmosphère ces particules interagissent de façon destructive avec les molécules d'air:
des particules secondaires (π^0, π^+, π^-, μ^+, μ^-, e^+, e^-) sont créées.

La majeure partie des particules détectables au niveau du sol sont des <u>muons</u>.



Une cascade de particules:
<u>une gerbe atmosphérique (an air shower)</u>

$\pi^+ \rightarrow \mu^+ + \nu_\mu$
$\quad \llcorner\!\rightarrow e^+ + \overline{\nu}_\mu + \nu_e$

Mais comment peut-il être possible de détecter un muon au niveau du sol?

Selon la Théorie de la Relativité, sa vitesse ***v*** ne peut pas dépasser la vitesse de la lumière ***c***

Même si le muon se déplace presque à la vitesse de la lumière, la distance parcourue par le muon est

$\boldsymbol{L = t \cdot 0.99\, c} = 2{,}2 \cdot 10^{-6}\,\mathrm{s} \cdot 0.99 \cdot 3 \cdot 10^{8}\,\mathrm{m/s} = \boldsymbol{653\ m}$

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Mais en réalité, les muons passent plusieurs kilomètres dans l'atmosphère... ?

Selon la Théorie de la Relativité, quand un muon se déplace avec une vitesse proche de celle de la lumière, la **contraction de la longueur** se produit:

$$\gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}$$

ce qui est plusieurs kilomètres <u>pour nous</u> est = juste quelques centaines de mètres <u>dans le cadre de référence du muon.</u>

$$\gamma = \frac{1}{\sqrt{1 - 0{,}99^2}} \approx 7$$

L’ = L · γ = 660 m · 7 ≈ 4.5 km

Le facteur de Lorentz

L'idée de l'expérience: Mesure de la vitesse des muons cosmiques

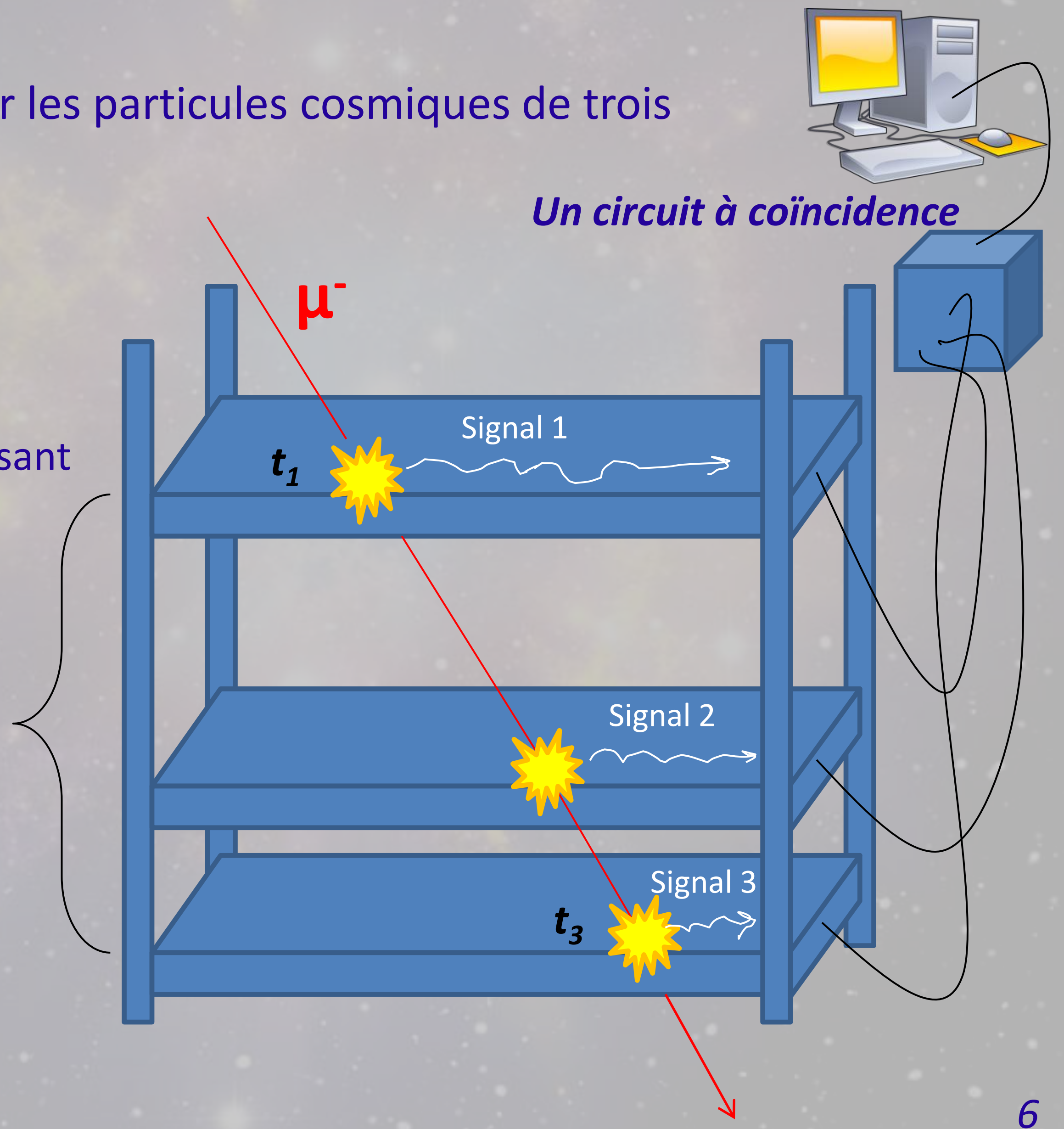
Nous disposons pour détecter les particules cosmiques de trois **scintillateurs** plastiques

Un scintillateur est un matériau qui émet de la lumière à la suite de l'absorption d'un rayonnement ionisant (photon ou particule chargée).

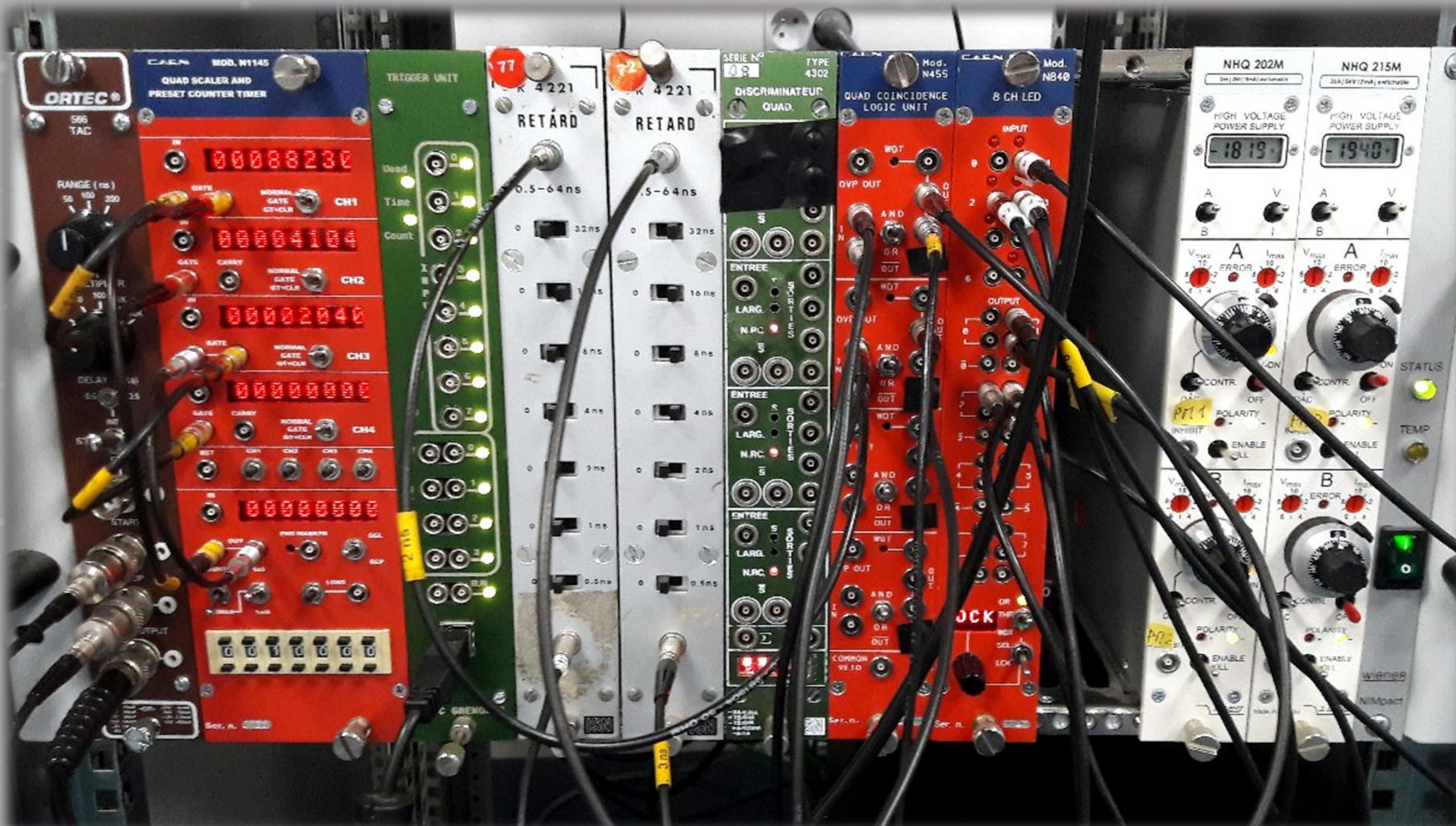
$\Delta t = t_3 - t_1$

La vitesse des muons:

$v = d / \Delta t$



Mais en fait tout cela est un petit peu plus compliqué...



Let's start, we'll figure out the rest on the way

Power supply

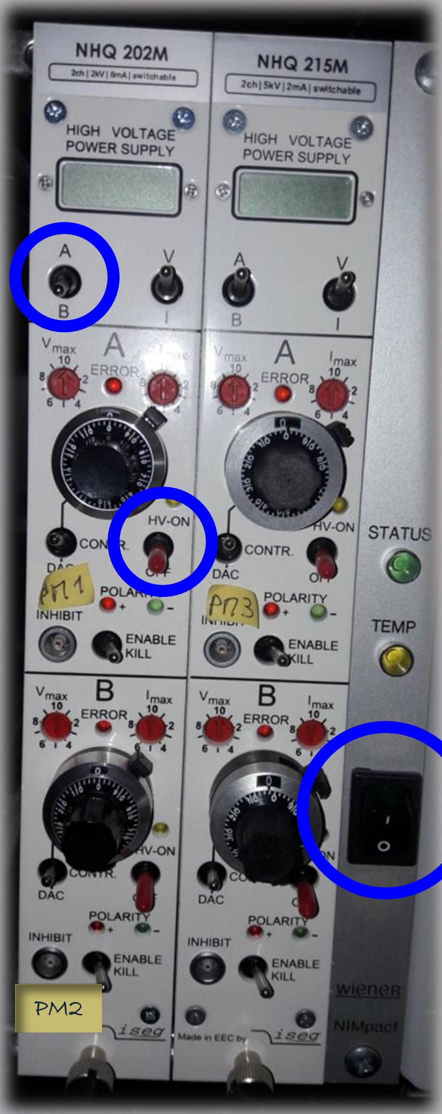
1. **Turn ON** the green button

2. **Set the high voltage** on the photomultipliers PM1, PM2, PM3:

    - *mode* ***«HV-ON»***

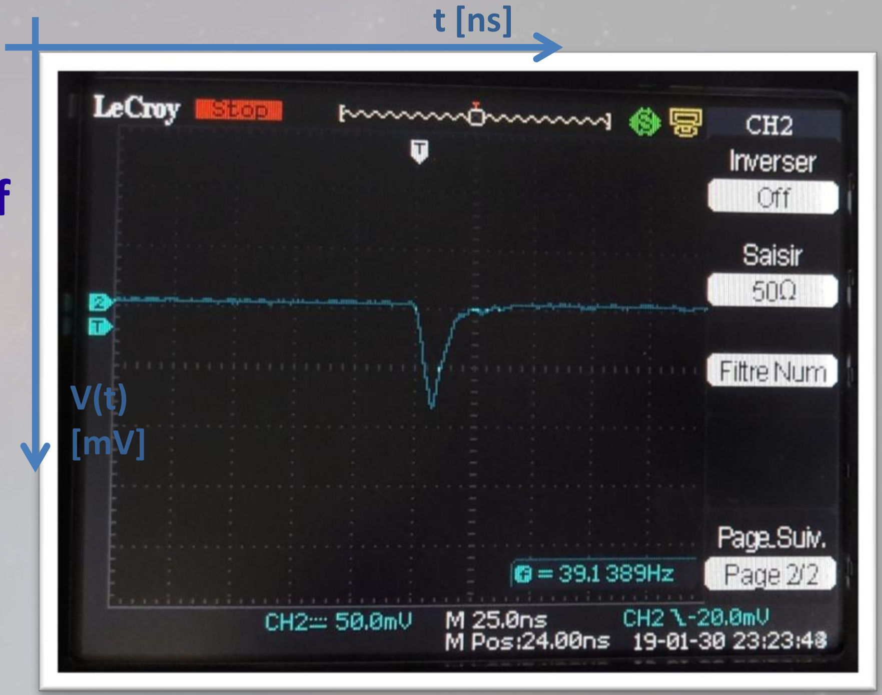
    - ***Voltage values*** *– see on the sheet*

    - *toggle switch:* ***A/B*** *– to visualize the values for PM1 or PM2 on the display*



Oscilloscope

Connect one of the PMs to an input of the oscilloscope and visualize the analog signal



Type of the analog signal:

a negative electric impulse of an average amplitude –100 mV and width $t \approx 25ns$.

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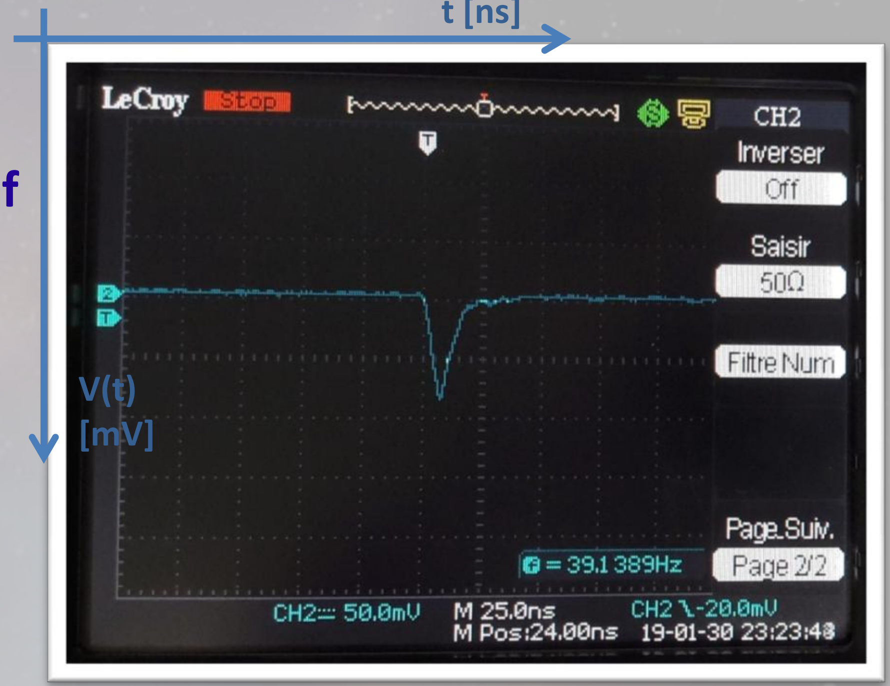

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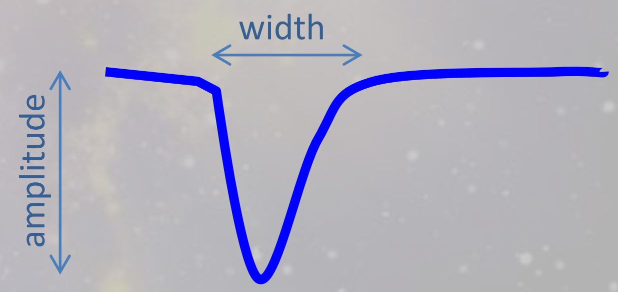
This analog signal can't be used by a computer.
An appropriate computer interpretation of it – is a **step signal** of a width comparable with the original signal width



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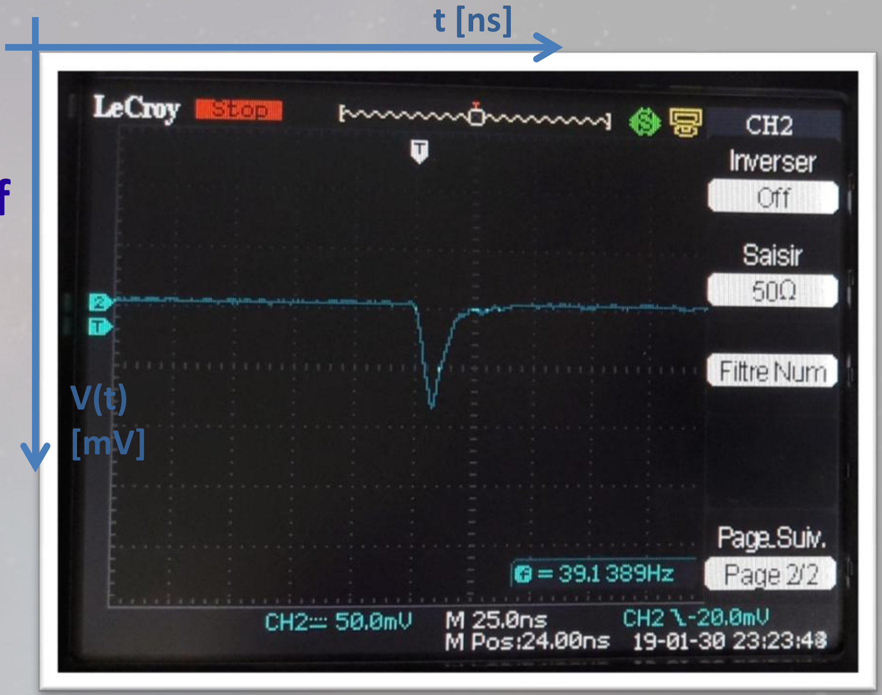

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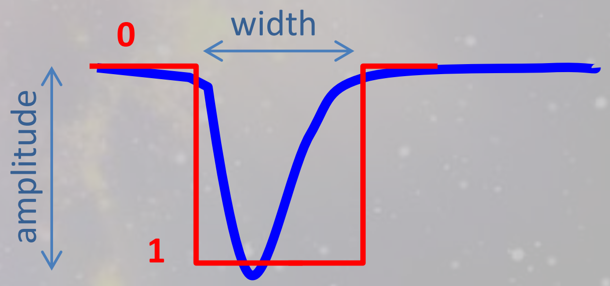
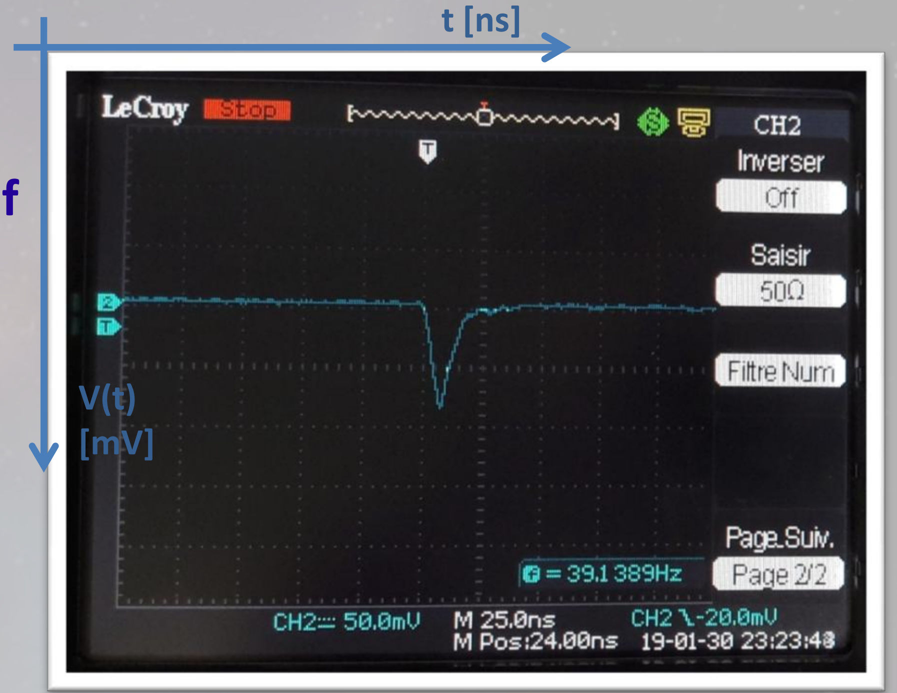


The corresponding step signal is generated by the **discriminator**

Oscilloscope



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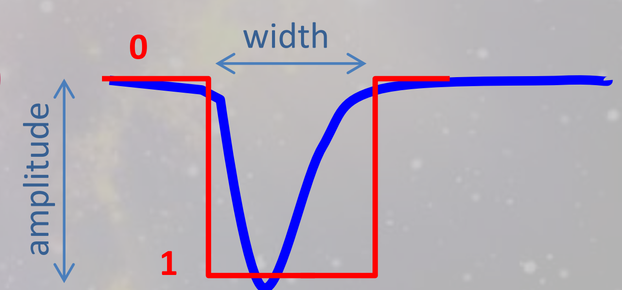
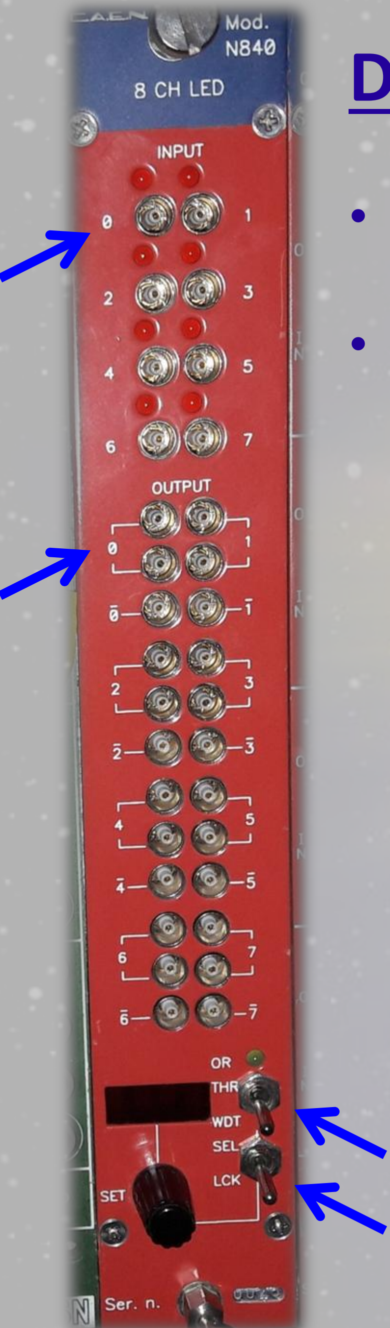
An appropriate computer interpretation of it – is a **step signal** of a width comparable with the original signal width
(if smaller => one signal can be misinterpreted as many)



The corresponding step signal is generated by the **discriminator**

Discriminator

- reads the signal from a PM and generates the corresponding step signal
- serves as a threshold



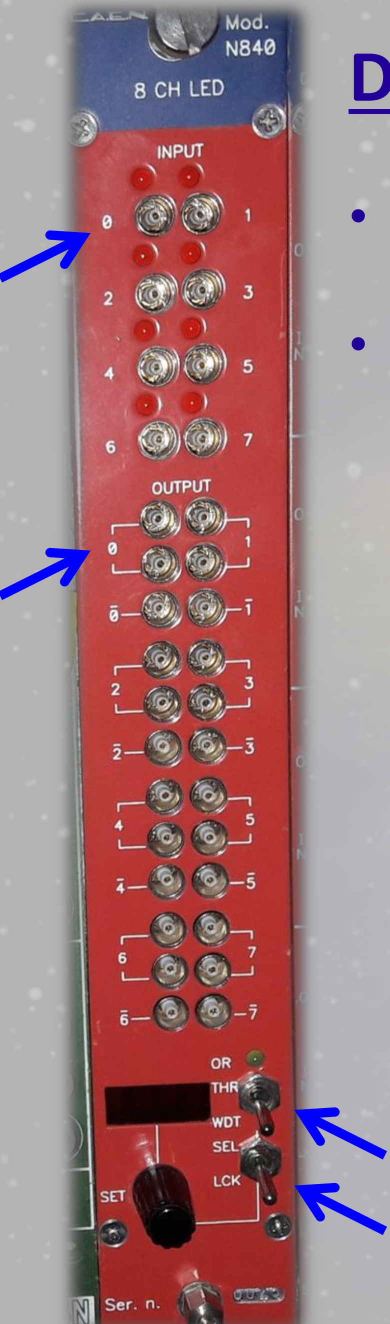
Discriminator

- **reads the signal from a PM and generates the corresponding step signal**
- **serves as a threshold**

1. Connect one of the PMs to the **INPUT(0)** of the discriminator
2. Take a cable and connect the **OUTPUT(0)** to the oscilloscope
3. Check the new signal on the oscilloscope. Does it have an adequate width?

SEL – to switch between different inputs
LCK – to lock (to fixate) the chosen value

Set **TRH ≈ 30 mV** – threshold value => **LCK**
WDT ≈ 232 – width value



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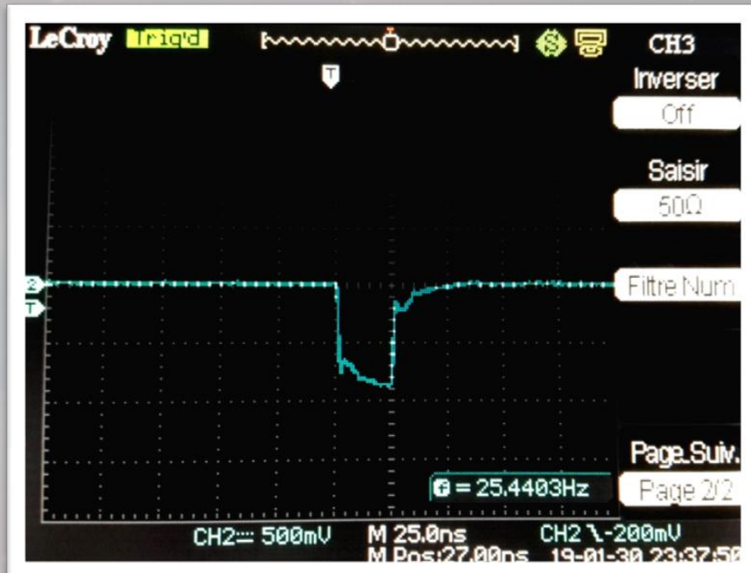

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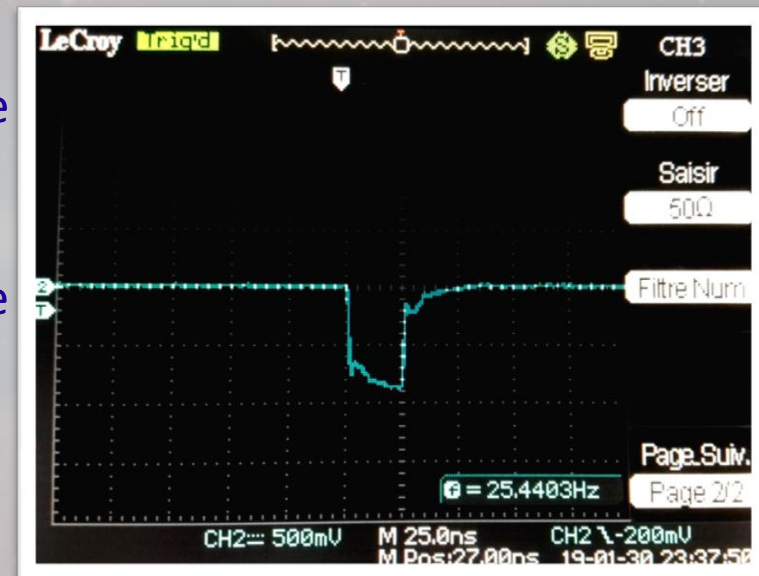

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Do this for all PM1, PM2, PM3

Now, when we have cut the noise, we know that the signal that we see corresponds to

a cosmic muon

It will go through all 3 detectors

Some radioactivity

from other experiments in the room, from the human body, etc.

Probability that it will go through all 3 detectors → 0

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<u>Coincidence scheme</u>

Logic circuit with 2 inputs

PM1: Yes

PM2: Yes

PM3: Yes

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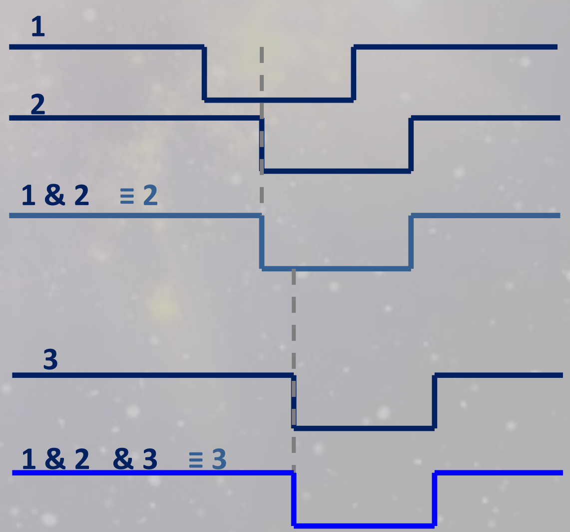
PM3: Yes

} **1 & 2 & 3**



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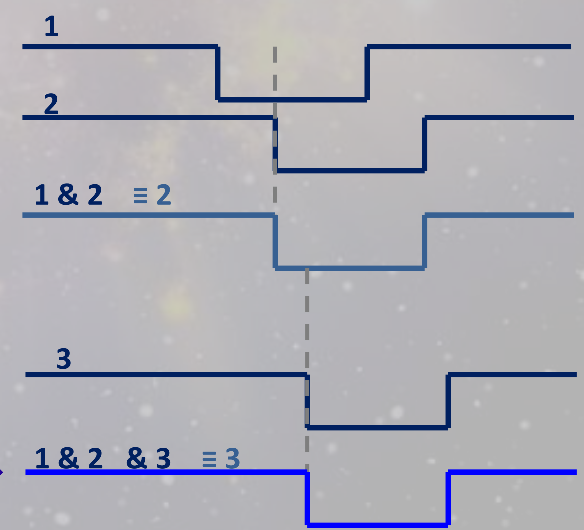

Only this signal is <u>for sure</u> from a muon ➡

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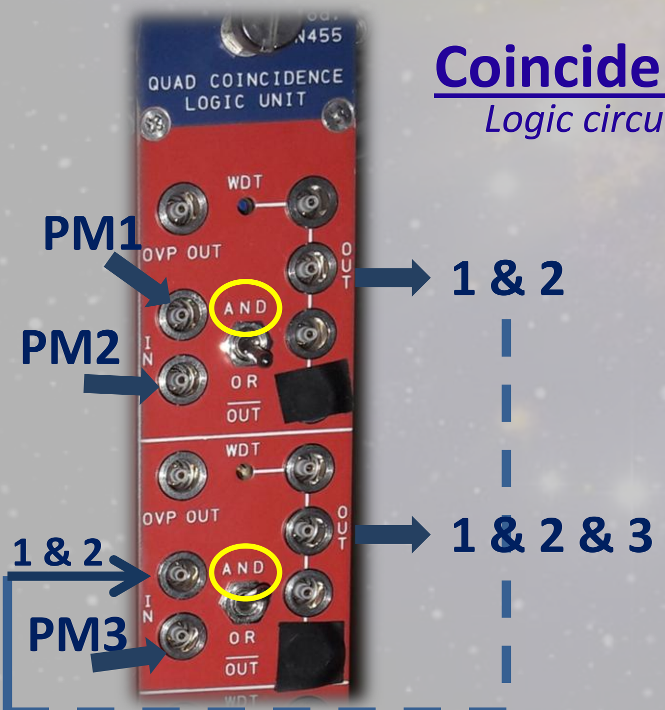

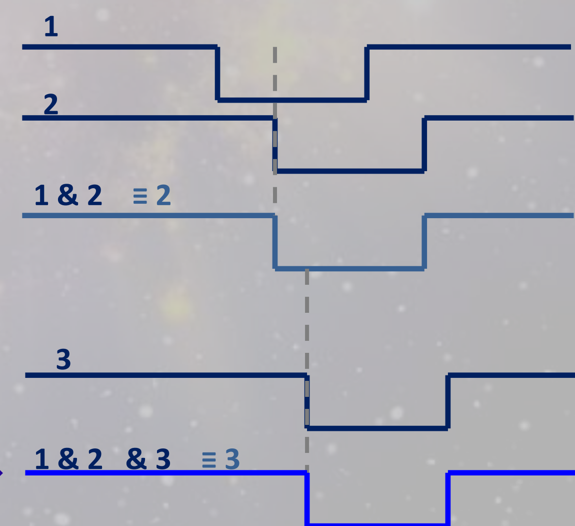

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In a perfect world

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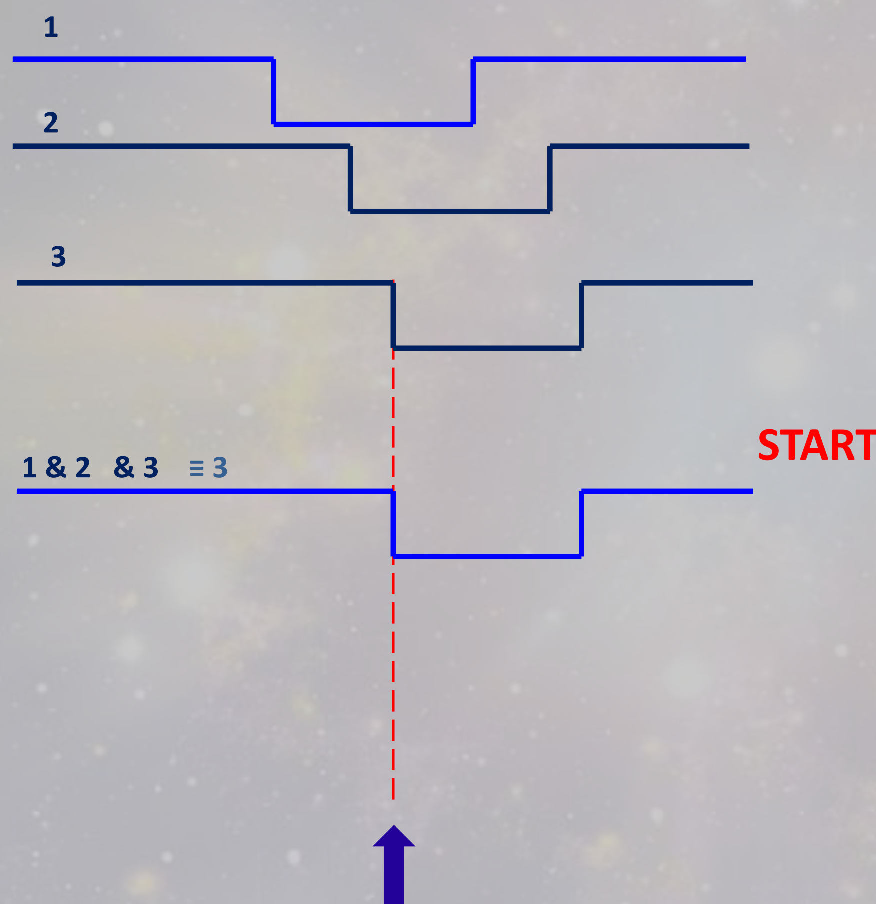
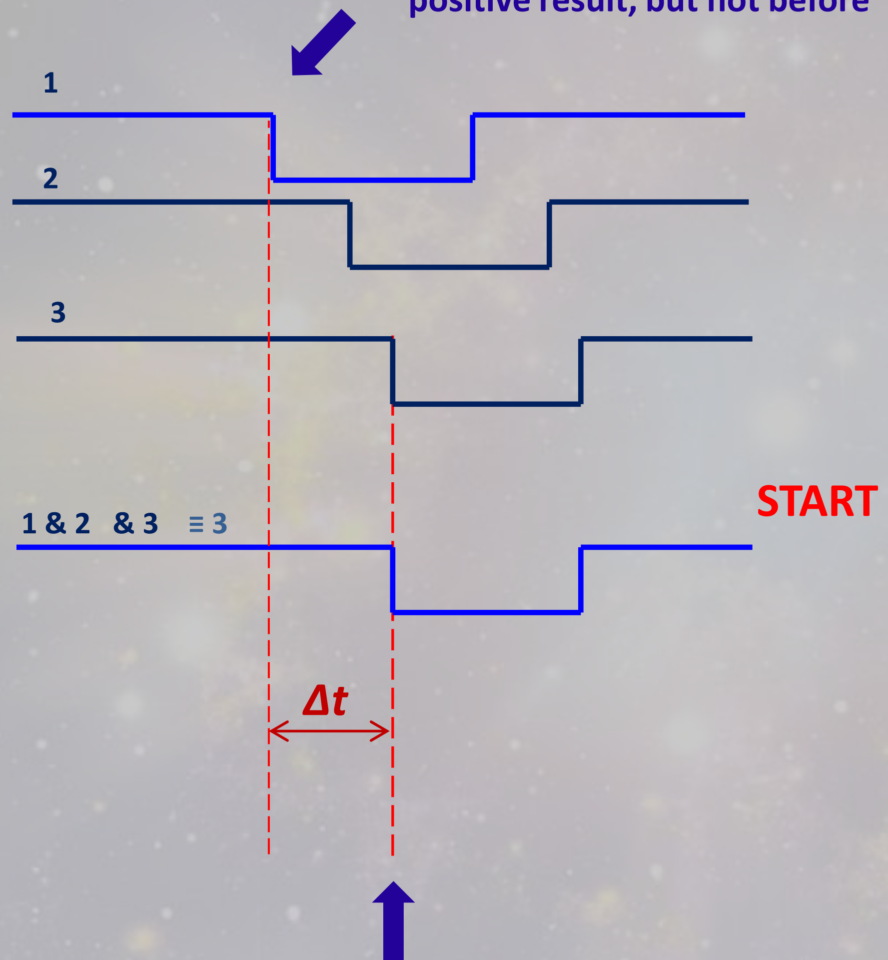


Only from this moment we know for sure that we have detected a muon, so this is the “START”

In a perfect world

But signal “1” is important only <u>after</u> the coincidence scheme gives a positive result, but not before



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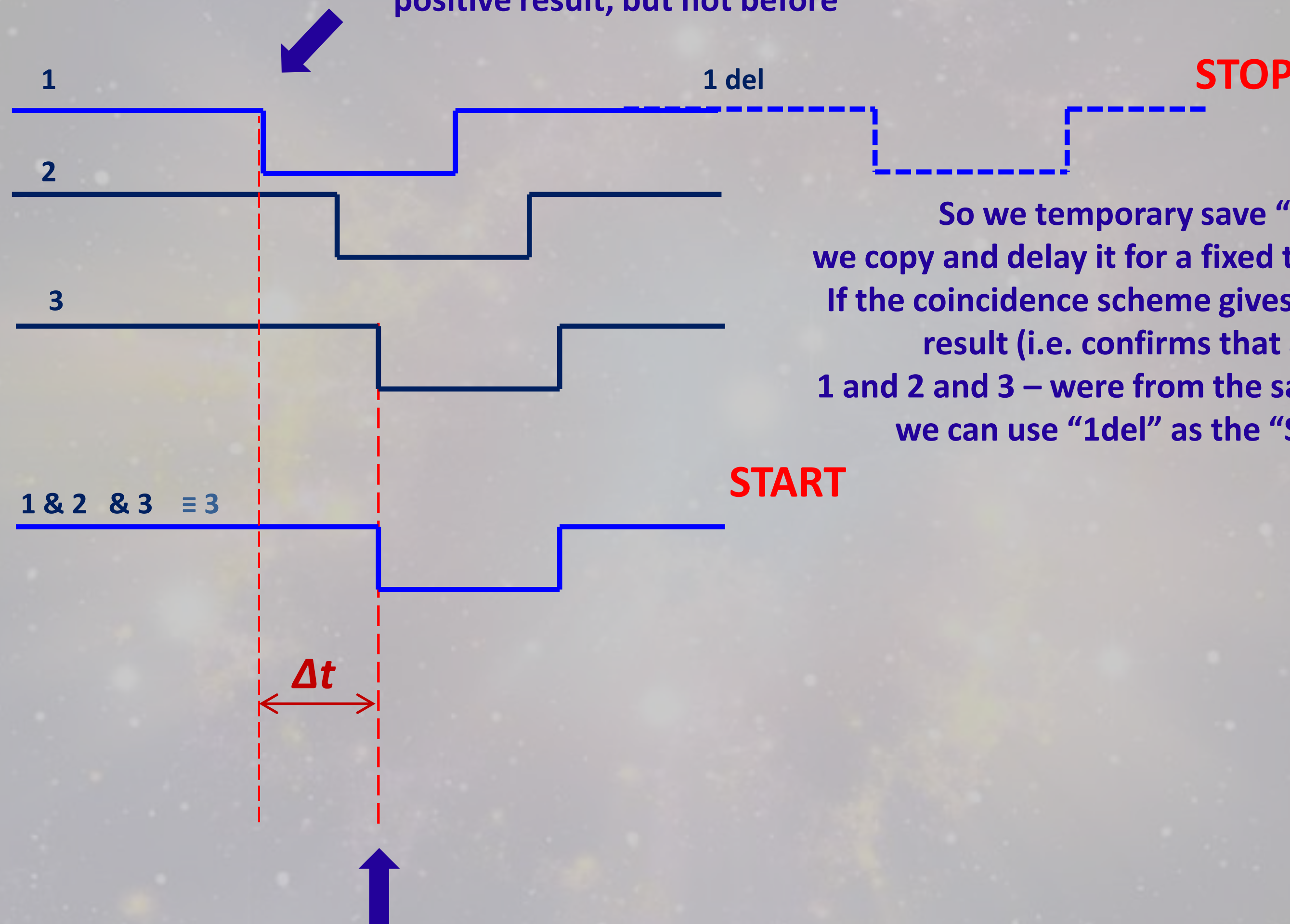

So we temporary save “1”:
we copy and delay it for a fixed time (64ns).
If the coincidence scheme gives a positive result (i.e. confirms that all –
1 and 2 and 3 – were from the same muon)
we can use “1del” as the “STOP”

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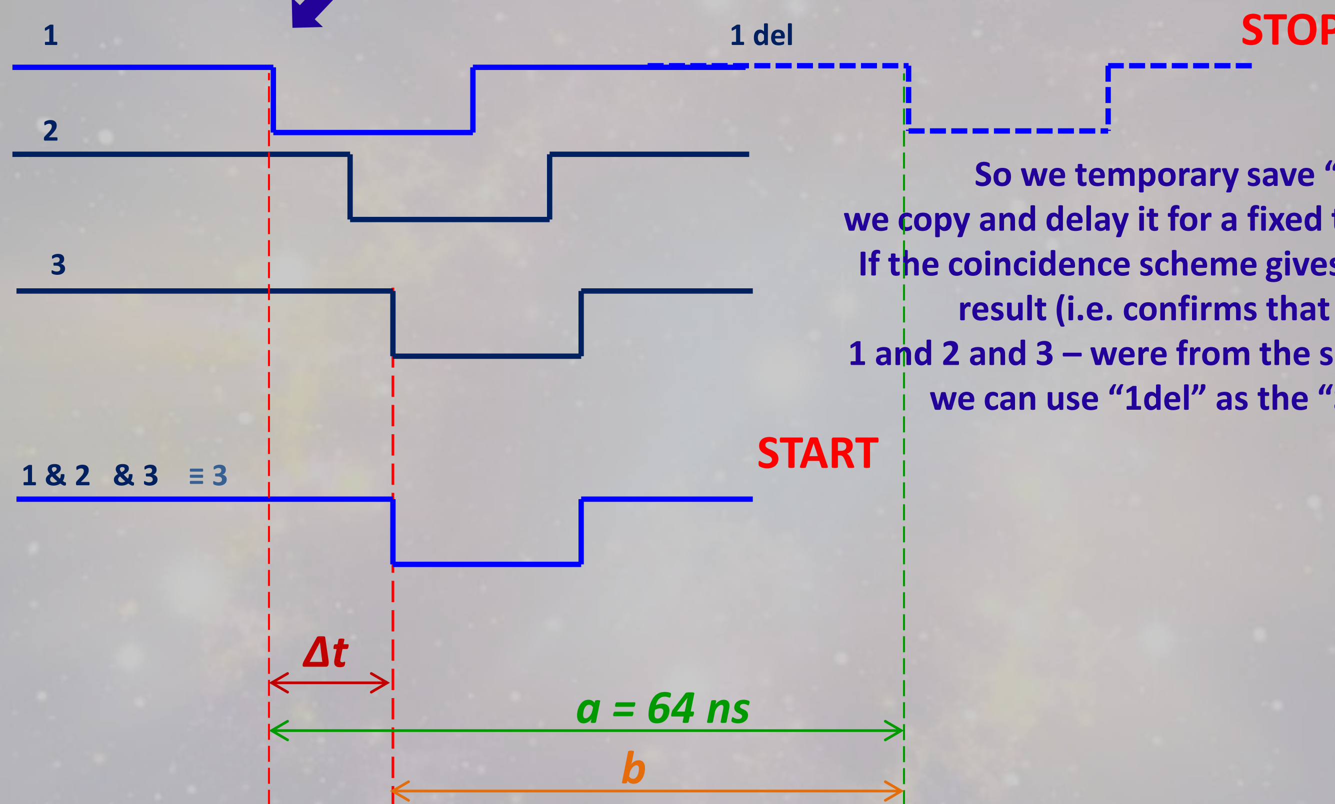

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$$\Delta t = a - b$$
$$= t_{START} - t_{STOP}$$

In a perfect world

IN/OUT

Options for the delay in [ns]

IN/OUT

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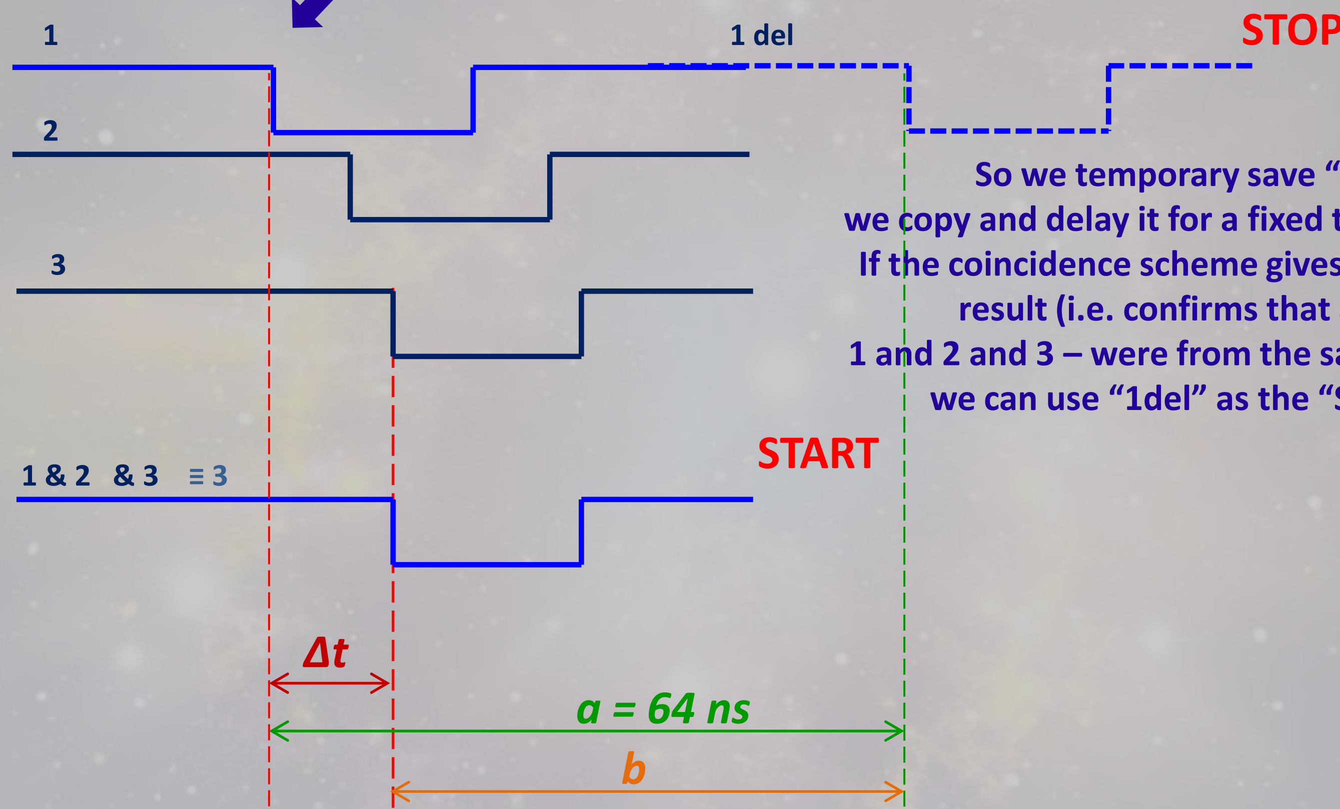

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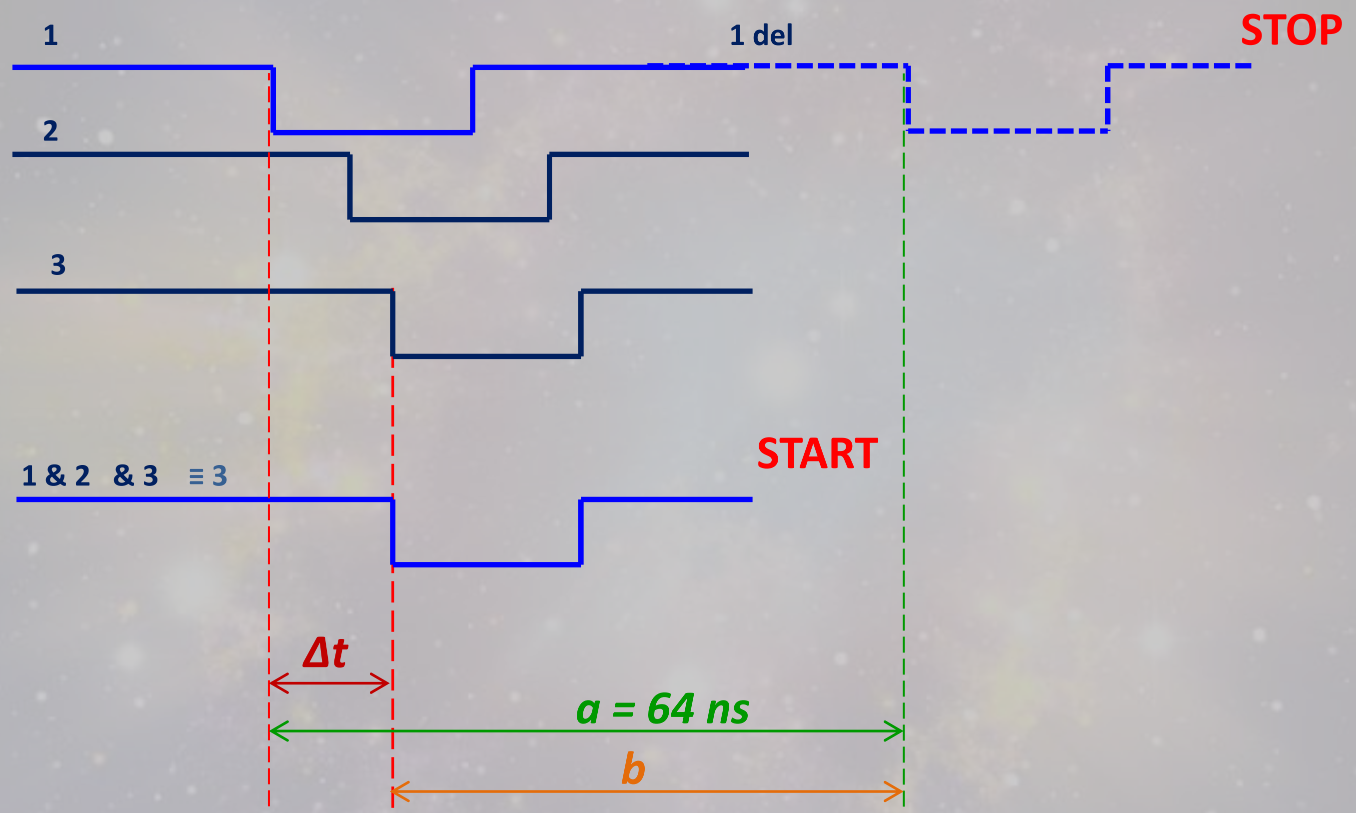
In REAL WORLD

Signals 1, 2 and 3 do not necessary appear in this order due to the different efficiency of the detectors

(e.g. sometimes "3" appears earlier that "2", sometimes not => the step signal on the oscilloscope will be oscillating)



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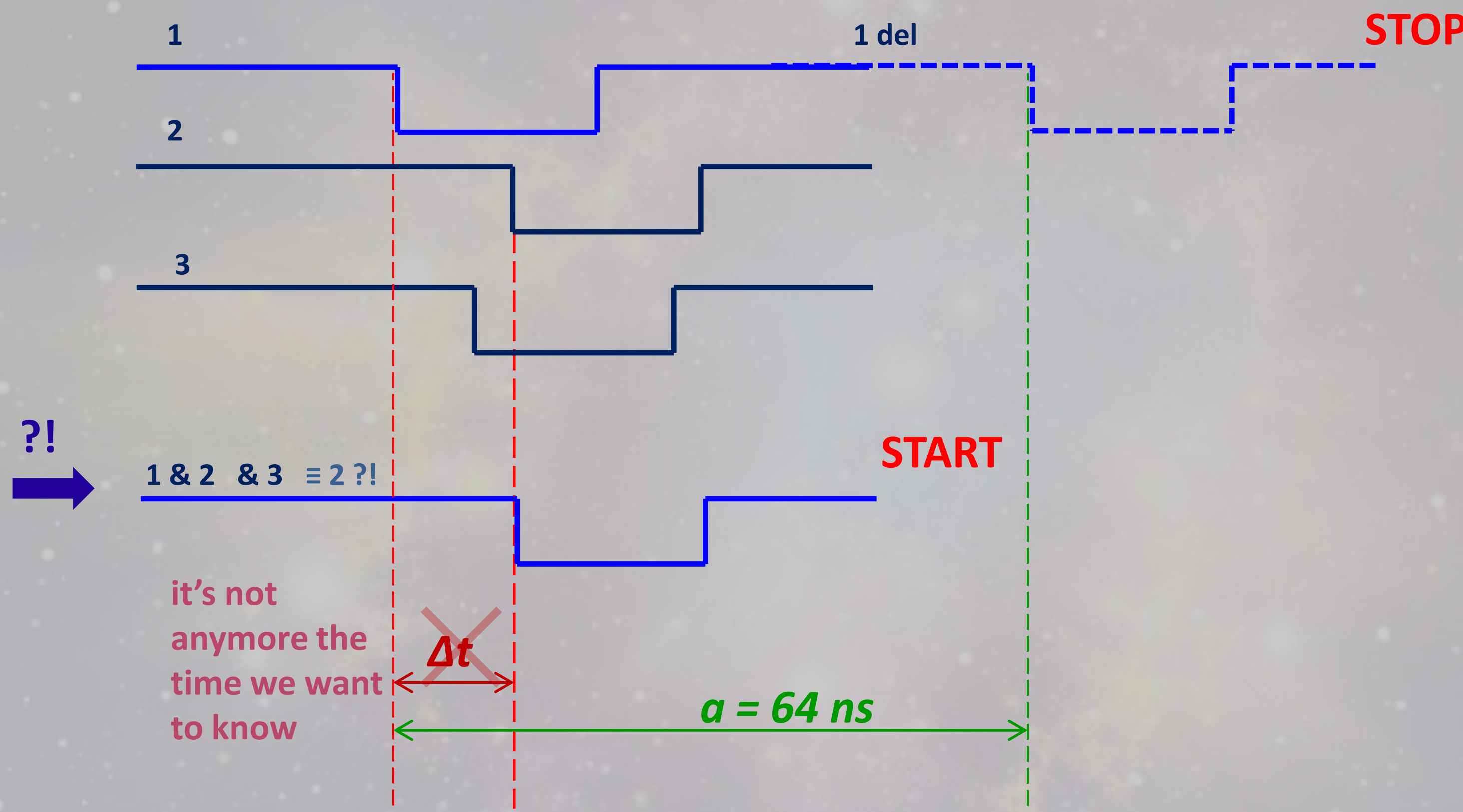

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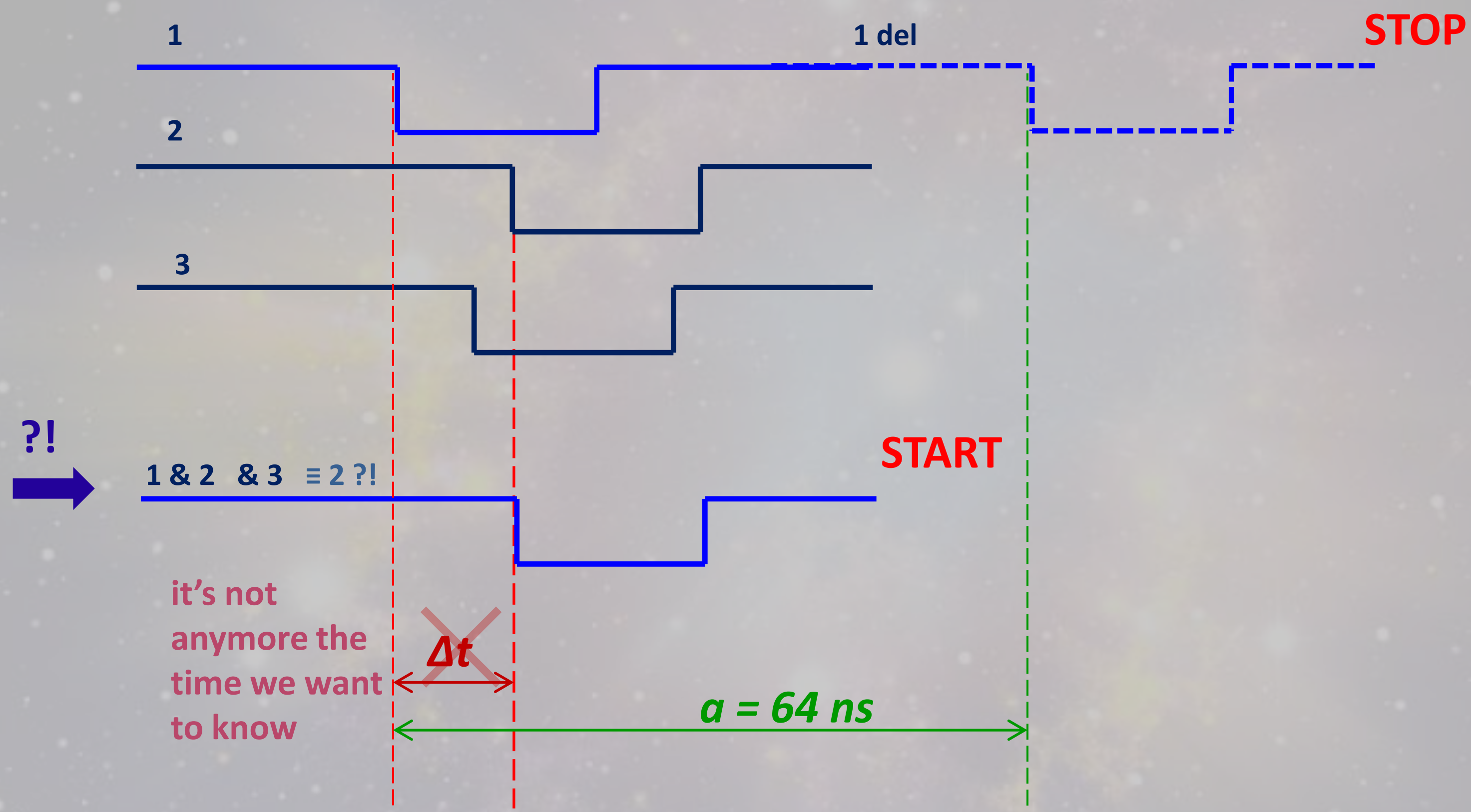

To measure the *Δt* between "1" *(actually "1del")* **and 1&2&3*(≡3)* we have to do so that the signal "3" would appear <u>the last</u>**

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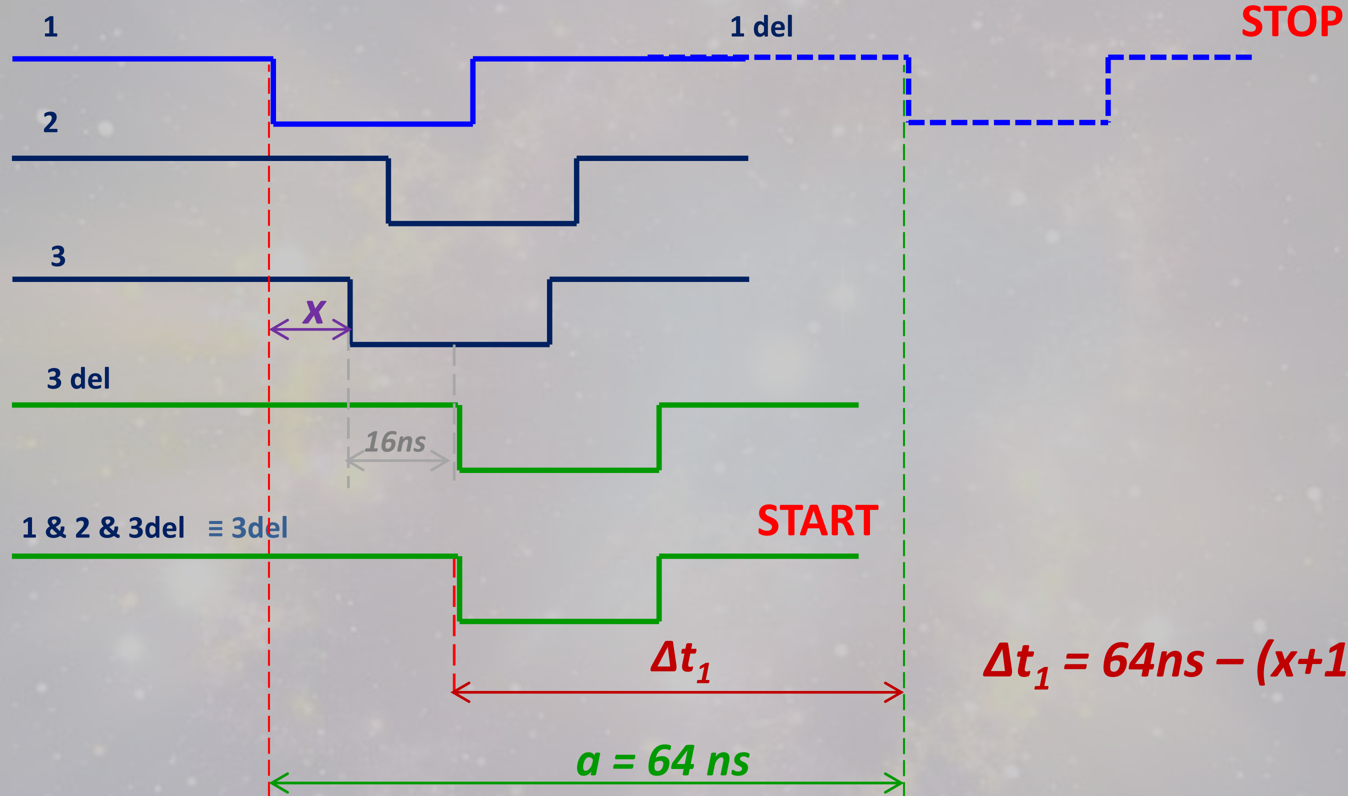


$\Delta t_1 = 64ns - (x+16ns)$

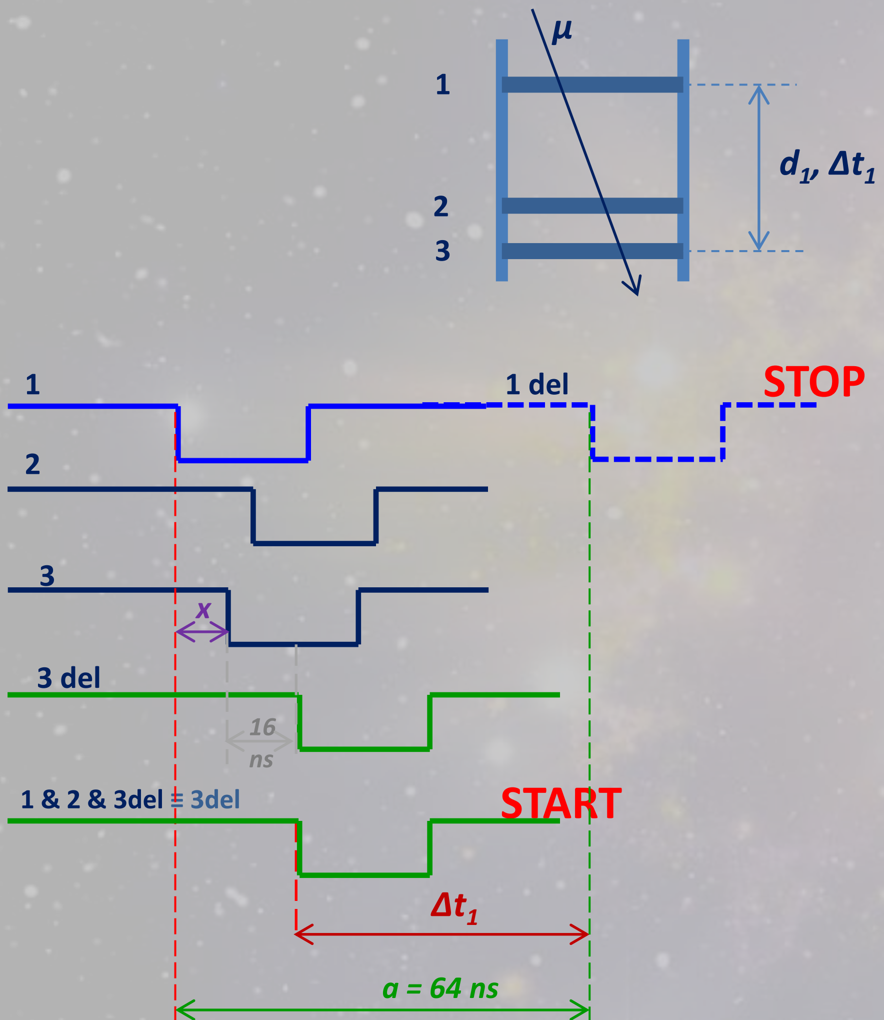
To measure the *Δt* between "1" *(actually "1del")* **and 1&2&3*(≡3)* we have to do so that the signal "3" would appear the last**

We delay the "3"

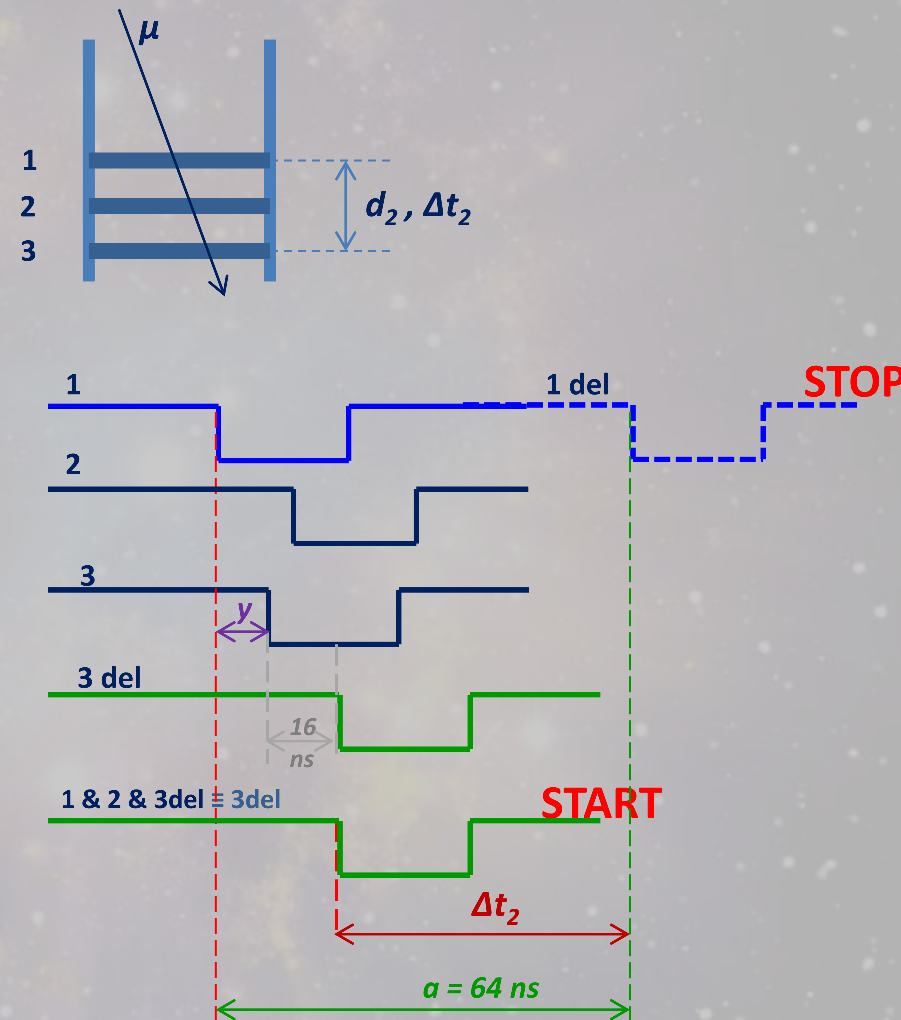
e.g. for 16ns

In REAL WORLD

Measurement 1: upper position



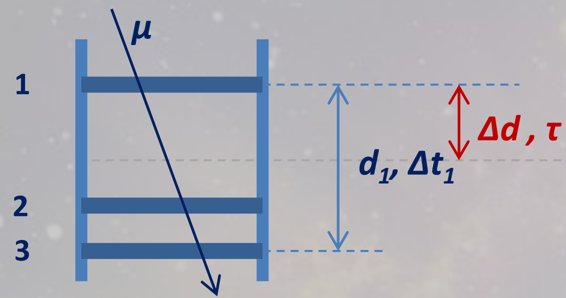
Measurement 2: lower position



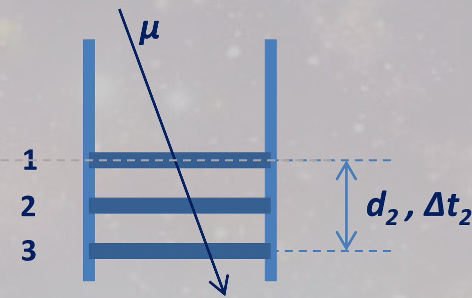
$$\Delta t_1 = 64ns - (x+16ns)$$
$$\Delta t_2 = 64ns - (y+16ns)$$

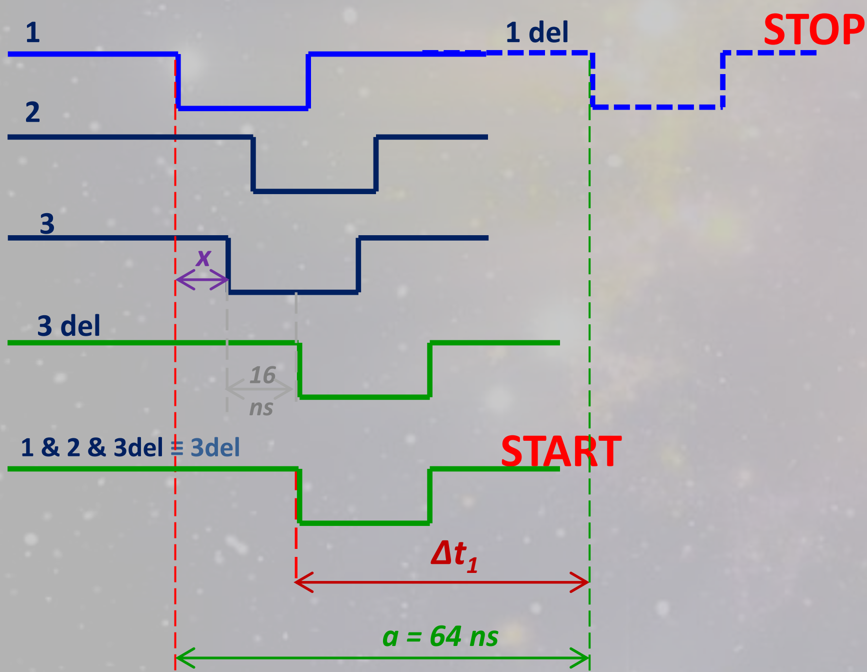
In REAL WORLD

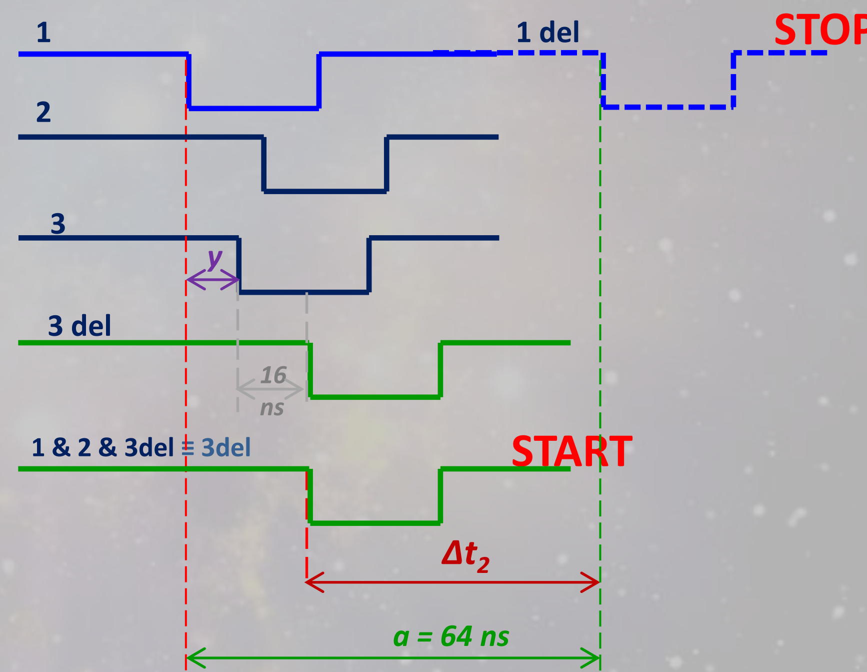
Measurement 1: upper position



Measurement 2: lower position







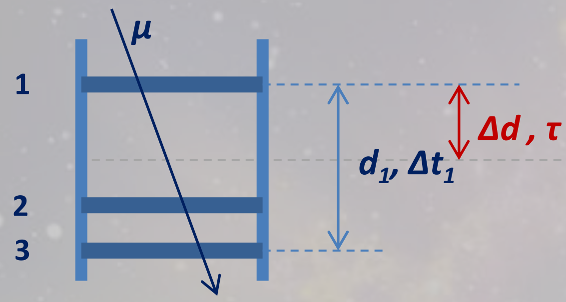
$$\Delta t_1 = 64ns - (x+16ns)$$
$$\Delta t_2 = 64ns - (y+16ns)$$

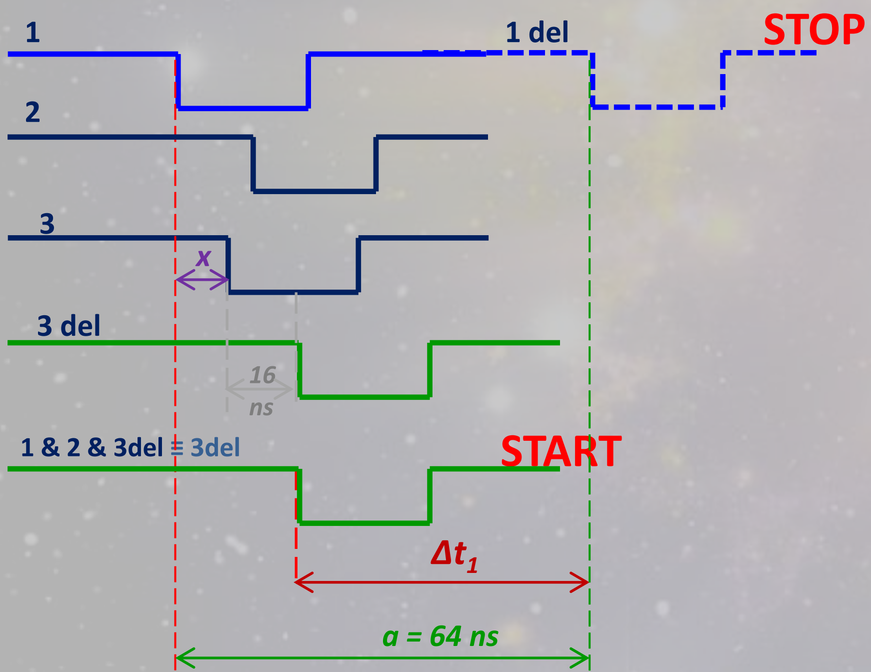
$$\Delta t_2 - \Delta t_1 = x - y = \tau$$

$$\boxed{v = \Delta d / \tau}$$

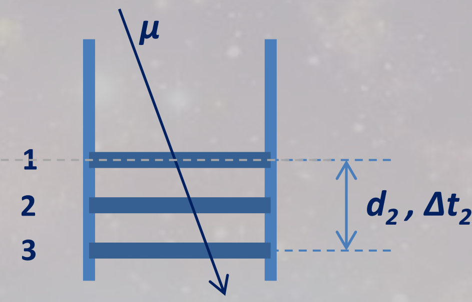
In REAL WORLD

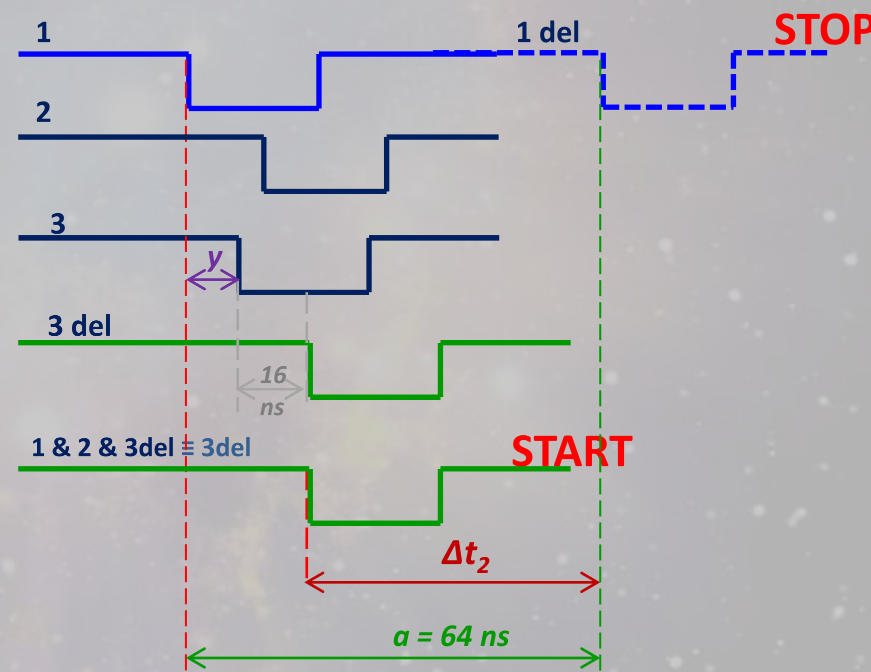
Measurement 1: upper position





Measurement 2: lower position





$$\Delta t_1 = 64ns - (x+16ns)$$
$$\Delta t_2 = 64ns - (y+16ns)$$

$$\Delta t_2 - \Delta t_1 = x - y = \tau$$

? *Why $\Delta t_2 > \Delta t_1$?*

$$\boxed{v = \Delta d / \tau}$$

TAC

(Time to Amplitude Converter)

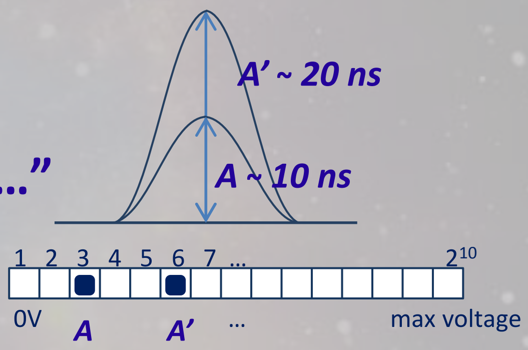
START

STOP

OUTPUT
=> PC

- **Measures *Δt* between START and STOP and generates an amplitude *A ~ Δt***

- ***The Multi-Channel Analyzer***
reports to the PC:
"a signal is received in the channel № …"

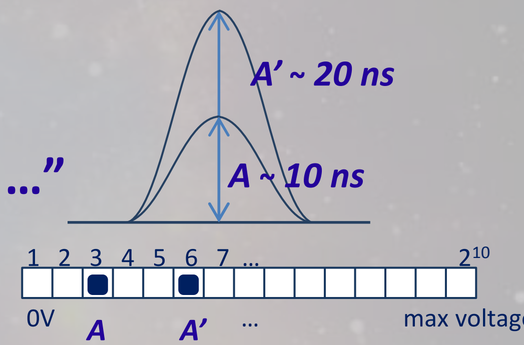


TAC

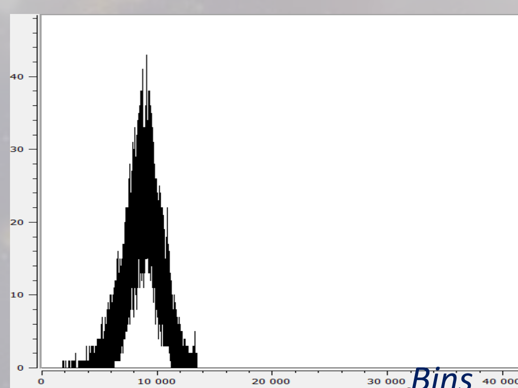
(Time to Amplitude Converter)

- **Measures *Δt* between START and STOP and generates an amplitude *A ~ Δt***

- ***The Multi-Channel Analyzer***
reports to the PC:
"a signal is received in the channel № …"



- **The PC builds a diagram:**



START

STOP

OUTPUT => PC

TAC

(Time to Amplitude Converter)

START

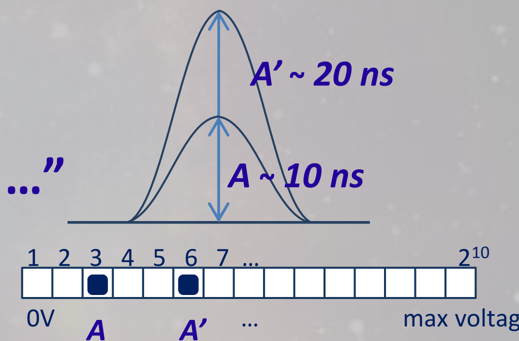
STOP

OUTPUT

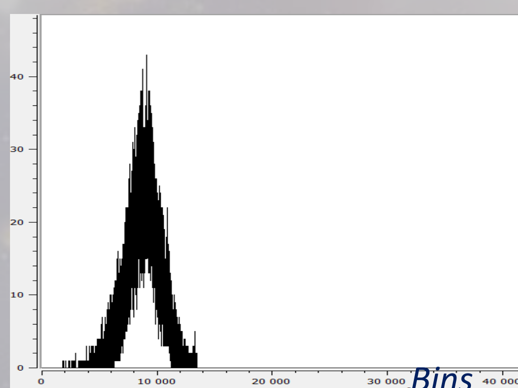
=> PC

- **Measures Δt between START and STOP and generates an amplitude $A \sim \Delta t$**

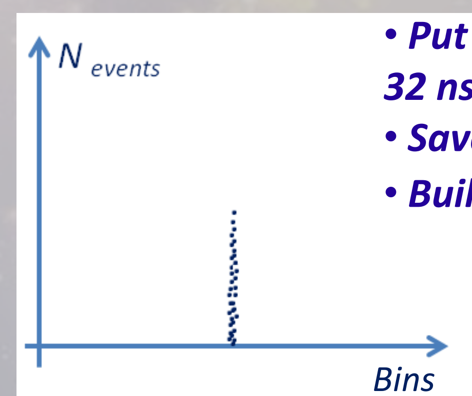
- ***The Multi-Channel Analyzer* reports to the PC: "a signal is received in the channel № …"**



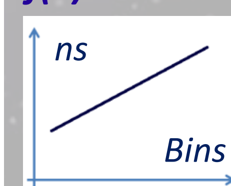
- **The PC builds a diagram:**



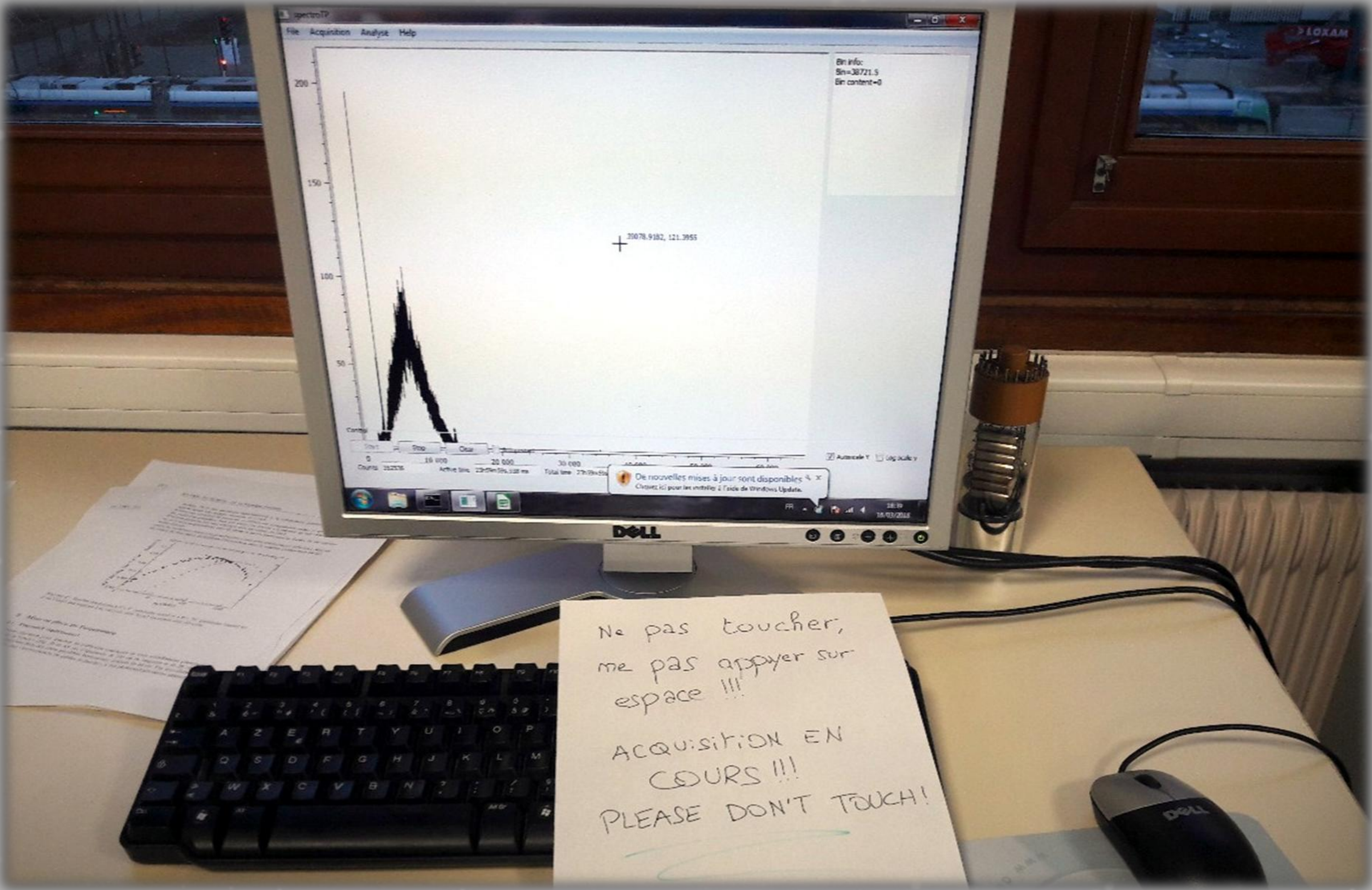
<u>Calibration of the TAC:</u>



- ***Put a delay (16 ns, 32 ns, 48 ns, 64 ns …)***
- ***Save the № of the bin***
- ***Build f(x)=bx+a***



Set Measurement 1, upper position (acquisition during 1 week)



Good luck!